

# Derived Intersection Theory on Foliation

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In this paper, I'll discuss foliation from point of view of derived algebraic geometry. Derived algebraic geometry proves the  $\infty$ -equivalence of dg Lie algebroid and foliation. Now the quotient is defined by quotient of dg  $L_\infty$  groupoid generated from dg Lie algebroid, and it's calculated by dg  $L_\infty$  version of Chevalley-Eilenberg complex. I'm also wondering evaluation of the foliation using language of DT/GW invariants. First of all, the idea of foliation is classically defined differential geometrically on a smooth manifold:

## Definition 0.1. Foliation

Foliation is a partition of a smooth manifold  $M$  into its submanifolds. One idea of generating foliation is to retrieve from distribution  $D_x \subset TM$  for some  $x \in M$ , and this distribution classifies the points  $x \in M$  to define foliation. Distribution  $D_x$  is defined by its vector field  $X, Y \in \Gamma(D_x)$  satisfying  $[X, Y] \in \Gamma(D_x)$ , and in this case  $D_x$  is called integrable. This  $D_x$  corresponds to a submanifold  $L$ , since there's its restriction  $D_x|_L = T_x L$  by the unique maximal submanifold  $L \subset M$  s.t.  $x \in L$ . Note in fact  $D_x|_{L_y} = T_y L$  for all  $y \in L$ . Accordingly, the resulting set  $\{L_y\}$  for  $y \in M$  is the set of submanifolds  $L_y \subset M$  and disjoint union  $\coprod L_y = M$ .

To verbalize the notion of foliation is an equivalent notion with integrability by Frobenius theorem, where integrability is done by Lie algebra. If Lie algebra describes the global property (there's no "if" in math Toynbee says), the corresponding Lie group quotient of manifold  $[M/G]$  is a stack. It doesn't work, however. I'll use Lie algebroid to describe every point of the submanifold  $L \subset M$ , whose Lie groupoid  $\mathcal{G}$  quotient is a quotient stack, which describes foliation  $[M/\mathcal{G}]$ . The first step of foliation my plan is to modernize the definition of foliation by using quotient stack with Lie algebroid/groupoid.

## Definition 0.2. Groupoid

A groupoid is a category in which every morphisms are invertible. Denote the objects are  $Ob(\mathcal{C}) = G_0$ , and the morphisms are  $Mor(\mathcal{C}) = G_1$ , the double arrow  $G_1 \rightrightarrows G_0$ . For an arrow  $g : x \rightarrow y$ , there are source and target maps  $s, t : G_1 \rightarrow G_0$ , together with composition  $G_1 \times_{G_0} G_1 \rightarrow G_1$ , identity arrow and inverses. A group becomes a Lie groupoid if  $G_0$  and  $G_1$  are smooth manifolds, and all the maps are smooth.

There are many important examples of Lie groupoid. The trivial things (logically equivalent to Lie group but somehow interesting generalization) are Lie group, which is a Lie groupoid with one object so  $G \rightrightarrows *$ . Another example which is more geometric is that Lie group action  $G \times X \rightrightarrows X$  can be modeled by  $G_1 = G \times X$  and  $G_0 = X$ . By the context of foliation, Lie groupoid  $Hol(\mathcal{F})$  of the foliation  $\mathcal{F}$  is holonomy groupoid. All the above examples are important. Now we consider quotient stack by Lie groupoid, in particular by Lie group. The definition of quotient stack below is a moduli stack while keeping the group symmetry structure for group  $G$ , but groupoid quotient  $\mathcal{G}$  is somewhat more complicated:

## Definition 0.3. Quotient Stack

For an algebraic variety  $X$ , quotient stack by Lie group  $G$  is  $[X/G]$  is a functor  $[X/G] : (Spaces)^{op} \rightarrow Groupoids$ .

$$[X/G](S) = \{(P \rightarrow S, f : P \rightarrow X) | P \rightarrow S \text{ is a principal } G\text{-bundle, } f_* f^* = \text{id} \}$$

In particular if  $X = pt$ , the quotient stack is  $[pt/G] = BG$ , and  $BG(pt)$  is one-object groupoid whose automorphism group is  $G = Hom(*, *)$ .

Meanwhile, quotient stack by groupoid  $[U/R]$  for a groupoid  $R$  is a stackification  $[U/R] = St([U/_p R])$  of category fibered in groupoids  $p : U \rightarrow R$ . This works for all  $c \in R$ , there exists  $x \in U$  s.t.  $p(x) = c$ , and for all  $f : x \rightarrow y$ ,  $p(f) = id_c$ .

where I abbreviate what the stackification means in this introductory chapter for the moment. In any case, the important quotient stack is holonomy groupoid quotient  $[M/Hol(F)]$ , which is a foliation. But how does Lie groupoid appears, where Lie algebra solves integrability condition in manifold? Lie algebra can be generalised to Lie algebroid by its globalization, where Lie groupoid comes. The definition of Lie algebroid is:

#### Definition 0.4. Lie Algebroid

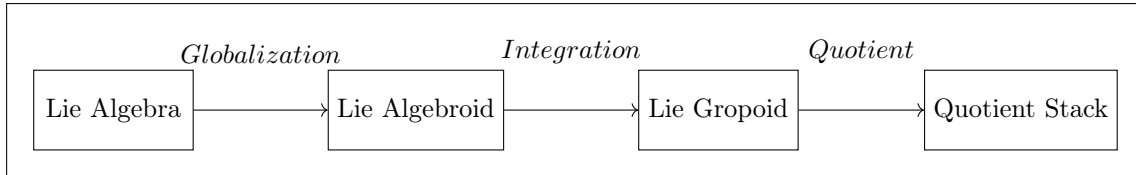
For a smooth manifold  $M$ , a Lie algebroid over  $M$  consists of  $(A \rightarrow M, [\cdot, \cdot]_A, \rho)$  where  $A \rightarrow M$  is a vector bundle,  $[\cdot, \cdot]_A : \Gamma(A) \times \Gamma(A) \rightarrow \Gamma(A)$  is a Lie bracket of sections,  $\rho : A \rightarrow TM$  is an anchor map. These satisfy product rule  $[X, fY]_A = f[X, Y]_A + \rho(X)(f)Y$ , and the anchor preserves bracket  $\rho([X, Y]_A) = [\rho(X), \rho(Y)]$ .

This Lie algebroid is a generalization of Lie algebra. In fact, the fiber  $A_x$   $x \in M$  of vector bundle  $A \rightarrow M$  is Lie algebra, so containing the data of local symmetry.

#### Remark 0.1

As a comparison, quotient stack  $[M/G]$  of gauge theory is a Lie group quotient, not Lie groupoid, where the local symmetry of particle action is everywhere the same, but the construction is nothing to do with the foliation problem. In other words, not all quotient stack is foliation. The computation of quotient stack  $[M/G]$  is done by Chevalley-Eilenberg complex. There's generalization of Chevalley-Eilenberg for the case of quotient stack by Lie groupoid.

In summary, the above argument can be depicted in the following flow chart, and now that the foliation is modernized geometrically, studying foliation is somewhat equivalent to studying quotient stack:



Several questions appear from this informal writing of foliation. First of all, to make the argument more precisely, can Lie algebroid be always integrated to the Lie groupoid? Now true in general, but Crainic-Fernandes integrability criterion claims Lie algebroid used in foliation can be always integrated.

#### Proposition 0.1. Crainic-Fernandes Criterion

TBD

Now let's set some research questions. To understand its structure, I consider beginning with specific geometry. Subjectively, groupoid foliation is too broad, think of Lie group quotient in gauge theory as a comparison, Lie groupoid quotient is much more dirty than that of Lie group, so making the shape  $M$  as simple as possible, for instance, to make it clear, and possibly  $M$  could be just projective space  $\mathbb{P}^n$  or a plane  $A^n$  or a hyperkähler manifold, I'd also study its computation using Chevalley-Eilenberg, Donaldson-Thomas/Gromov-Witten invariants or any other, while it's interesting tracking what's the missing data when we define quotient stack of Lie groupoid. On the other hand, we could also consider the categorification to make it more abstract. Typically, I noticed derived algebraic geometry might be useful to discuss these problems, and I'll consider geometrization of moduli stack of foliations where we can discuss intersection or any others:

### Question 0.1

- Making geometry  $M$  as simple as possible?
- Computation of groupoid version of CE cohomology? DT/GW invariants?
- Does quotient stack contain all the necessary information?
- Is there either deformation or quantization or derived geometry?
- Categorification?

Another view point is that geometry is useful in QFT, for example. Especially, consider Lagrangian intersection theory of Calabi-Yau geometry, whose quantum cohomology generates Gromov-Witten invariant, and this helps to compute path integral in both 4d/10d gauge theory. For this time I'll introduce derived Lagrangian intersection theory to this foliation. Suppose there are two distinct Lagrangian foliations  $\mathcal{F}_1$  and  $\mathcal{F}_2$  on the same geometry  $M$ , and if considered intersection of these foliations, this might be interesting and different.

Furthermore, if the whole argument contains  $dgL_\infty$  algebroid as generalization of Lie algebroid, I wonder if there is any comparison with  $A_\infty$  side in particular Fukaya category, or if there's any connection to homological mirror symmetry? I'll simply summarize in the list:

### Question 0.2

- What happens if the foliation is Lagrangian?
- Is there any connection to Fukaya category?

For the organization of this paper, I begin with language of derived algebraic geometry and  $\infty$  categories, then enhancing groupoid foliation in derived language as well as classically, and I add Lagrangian intersection theory.

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## 1 Derived Algebraic Geometry

Derived scheme stands with  $\infty$ -functor  $F : cdga \rightarrow \infty - groupoid$ . I begin with defining  $\infty$ -category.

### Definition 1.1. Derived Affine Scheme

### Definition 1.2

## 2 Groupoid Foliation

### Definition 2.1

### Definition 2.2

## 3 Quotient Stack

GIT?

## 4 Lagrangian Intersection Theory

The question is if there is any intersection theory of two foliations of Lagrangians.

### Definition 4.1. Lagrangian

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